

Syllabus - B.Tech CSE S5

MES Institute of Technology and Management

Official Document

CST301 - Theory of Computation

Module 1: Finite Automata

Introduction to Automata Theory, Strings, Languages.

DFA (Deterministic Finite Automata): Definition, Transition diagram.

NFA (Nondeterministic Finite Automata): Definition, Equivalence of NFA and DFA.

Module 2: Regular Expressions

Regular Expressions and Languages.

Pumping Lemma for Regular Languages.

Closure properties of Regular Languages.

Assessment

Internal Marks: 50 (2 Series Exams + Assignments)

End Semester Exam: 100

Credits: 4

CST306 - Algorithm Analysis and Design

Module 1: Complexity Analysis

Time and Space Complexity. Big-O notation.

Divide and Conquer: Merge Sort, Quick Sort.

Module 2: Dynamic Programming

Principle of Optimality.

Knapsack Problem, Longest Common Subsequence.

Faculty

Course Coordinator: Dr. Anitha S (HOD CSE)